

# Valgrind 检测

## 测试说明

由于全量数据集（3319 个文本片段）在 Valgrind 下运行时间过长，本次测试使用包含 2 个文本片段的测试数据进行验证。

内存泄漏与数据量无关，小数据集足以验证代码路径上的内存管理是否正确。

下为检测程序输出片段，完整输出（包含截图）在文尾

## 程序运行输出

```
[INFO] 正在处理文档: ./data/test_tiny.txt
[INFO] 当前文档切分完成，共 2 个片段
[INFO] 全部文档切分完成，共 2 个片段
[INFO] 正在加载 Embedding 模型: ./models/bge-small-zh-v1.5-f16.gguf
[INFO] 正在生成向量嵌入...
[INFO] 嵌入进度: 2/2
[INFO] 正在写入向量数据库...
[INFO] 知识库构建完成，保存至: ./data/test_tiny.db
[INFO] 数据库包含 2 个向量条目
```

## Valgrind 检测结果

### HEAP SUMMARY (堆内存汇总)

```
==463887== HEAP SUMMARY:
==463887==      in use at exit: 126,456 bytes in 15 blocks
==463887==    total heap usage: 28,480 allocs, 28,465 frees, 46,596,058 bytes
allocated
```

## LEAK SUMMARY (泄漏汇总)

==463887== LEAK SUMMARY:

==463887== definitely lost: 0 bytes in 0 blocks

==463887== indirectly lost: 0 bytes in 0 blocks

==463887== possibly lost: 960 bytes in 3 blocks

==463887== still reachable: 125,496 bytes in 12 blocks

==463887== suppressed: 0 bytes in 0 blocks

## Linux 系统及 Valgrind 版本展示

```
labuser@labuser:~/CPP_Project$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
labuser@labuser:~/CPP_Project$ valgrind --version
valgrind-3.22.0
```

# 完整输出和截图附件

```
labuser@labuser:~/CPP_Project$ valgrind --leak-check=full --show-leak-kinds=all \
    ./build_linux/mrag --ingest ./data/test_tiny.txt ./data/test_tiny.db
==463887== Memcheck, a memory error detector
==463887== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==463887== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==463887== Command: ./build_linux/mrag --
ingest ./data/test_tiny.txt ./data/test_tiny.db
==463887==

[INFO] 正在处理文档: ./data/test_tiny.txt
[INFO] 当前文档切分完成, 共 2 个片段
[INFO] 全部文档切分完成, 共 2 个片段
[INFO] 正在加载 Embedding 模型: ./models/bge-small-zh-v1.5-f16.gguf
[INFO] 正在生成向量嵌入...
[INFO] 嵌入进度: 2/2
[INFO] 正在写入向量数据库...
[INFO] 知识库构建完成, 保存至: ./data/test_tiny.db
[INFO] 数据库包含 2 个向量条目
==463887==
==463887== HEAP SUMMARY:
==463887==      in use at exit: 126,456 bytes in 15 blocks
==463887==    total heap usage: 28,480 allocs, 28,465 frees, 46,596,058 bytes
allocated
==463887==
==463887== 8 bytes in 1 blocks are still reachable in loss record 1 of 13
==463887==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-
amd64-linux.so)
```

```

==463887==      by 0x5BEB4A4: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==      by 0x5C0210E: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==      by 0x5BE9C5B: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==      by 0x400571E: call_init.part.0 (dl-init.c:74)
==463887==      by 0x4005823: call_init (dl-init.c:120)
==463887==      by 0x4005823: _dl_init (dl-init.c:121)
==463887==      by 0x401F59F: ??? (in /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2)
==463887==      by 0x3: ???
==463887==      by 0x1FFEFFFFE02: ???
==463887==      by 0x1FFEFFFFE15: ???
==463887==      by 0x1FFEFFFFE1E: ???
==463887==      by 0x1FFEFFFFE33: ???
==463887==
==463887== 8 bytes in 1 blocks are still reachable in loss record 2 of 13
==463887==      at 0x4846FA3: operator new(unsigned long) (in
/usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==      by 0x528A530: ggml_backend_cpu_repack_buffer_type()
(repack.cpp:4832)
==463887==      by 0x5271F1A:
ggml_backend_cpu_get_extra_buffer_types()::{lambda()#1}::operator()() const (ggml-
cpu.cpp:65)
==463887==      by 0x5271FF5: ggml_backend_cpu_get_extra_buffer_types() (ggml-
cpu.cpp:71)
==463887==      by 0x52720AC:
ggml_backend_cpu_device_get_extra_buffers_type(ggml_backend_device*)::{lambda()#1
}::operator()() const (ggml-cpu.cpp:78)
==463887==      by 0x5272191:
ggml_backend_cpu_device_get_extra_buffers_type(ggml_backend_device*) (ggml-
cpu.cpp:81)

```

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==463887==    by 0x4EBE820: make_cpu_buf_list(std::vector<llama_device,
std::allocator<llama_device> > const&, bool, bool) (llama-model.cpp:861)

==463887==    by 0x4EC0DE9:
llama_model_base::load_tensors(llama_model_loader&) (llama-model.cpp:1178)

==463887==    by 0x4D5E6CE: llama_model_load(gguf_context*, void
*)(ggml_tensor*, void*), void*, std::__cxx11::basic_string<char, std::char_traits<char>,
std::allocator<char> > const&, std::vector<std::__cxx11::basic_string<char,
std::char_traits<char>, std::allocator<char> >,
std::allocator<std::__cxx11::basic_string<char, std::char_traits<char>,
std::allocator<char> > > &, _IO_FILE*, llama_model_params&) (llama.cpp:327)

==463887==    by 0x4D5EE4F: llama_model_load_from_file_impl(gguf_context*, void
*)(ggml_tensor*, void*), void*, std::__cxx11::basic_string<char, std::char_traits<char>,
std::allocator<char> > const&, std::vector<std::__cxx11::basic_string<char,
std::char_traits<char>, std::allocator<char> >,
std::allocator<std::__cxx11::basic_string<char, std::char_traits<char>,
std::allocator<char> > > &, _IO_FILE*, llama_model_params) (llama.cpp:386)

==463887==    by 0x4D5F250: llama_model_load_from_file (llama.cpp:427)

==463887==    by 0x13DF44:
LlamaModelBase::LlamaModelBase(std::__cxx11::basic_string<char,
std::char_traits<char>, std::allocator<char> > const&, ModelMode,
AppConfig::GenerationConfig const&, AppConfig::EmbeddingConfig const&, int)
(llamamodel.cc:18)

==463887==

==463887== 40 bytes in 1 blocks are still reachable in loss record 3 of 13

==463887==    at 0x484DB80: realloc (in
/usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)

==463887==    by 0x5BEB514: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)

==463887==    by 0x5C00A09: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)

==463887==    by 0x5BF5971: GOMP_parallel (in /usr/lib/x86_64-linux-
gnu/libgomp.so.1.0.0)

==463887==    by 0x5270C2E: ggml_graph_compute (ggml-cpu.c:3333)

==463887==    by 0x5272636: ggml_backend_cpu_graph_compute(ggml_backend*,

```

ggml\_cgraph\*) (ggml-cpu.cpp:190)

==463887== by 0x5545BE3: ggml\_backend\_graph\_compute\_async (ggml-backend.cpp:452)

==463887== by 0x554AD50: ggml\_backend\_sched\_compute\_splits(ggml\_backend\_sched\*) (ggml-backend.cpp:1678)

==463887== by 0x554BD1C: ggml\_backend\_sched\_graph\_compute\_async (ggml-backend.cpp:1901)

==463887== by 0x4DB0BDD: llama\_context::graph\_compute(ggml\_cgraph\*, bool) (llama-context.cpp:2326)

==463887== by 0x4DAB802: llama\_context::process\_ubatch(llama\_ubatch const&, llm\_graph\_type, llama\_memory\_context\_i\*, ggml\_status&) (llama-context.cpp:1309)

==463887== by 0x4DABC9A: llama\_context::encode(llama\_batch const&) (llama-context.cpp:1382)

==463887==

==463887== 96 bytes in 1 blocks are still reachable in loss record 4 of 13

==463887== at 0x484D953: calloc (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x5BEB4DC: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5BE9BA0: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5BE9C77: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x400571E: call\_init.part.0 (dl-init.c:74)

==463887== by 0x4005823: call\_init (dl-init.c:120)

==463887== by 0x4005823: \_dl\_init (dl-init.c:121)

==463887== by 0x401F59F: ??? (in /usr/lib/x86\_64-linux-gnu/ld-linux-x86-64.so.2)

==463887== by 0x3: ???

==463887== by 0x1FFEFFFFE02: ???

==463887== by 0x1FFEFFFFE15: ???

==463887== by 0x1FFEFFFFE1E: ???

==463887== by 0x1FFEFFFFE33: ???

==463887==

==463887== 192 bytes in 1 blocks are still reachable in loss record 5 of 13

==463887== at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x5BEB4A4: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5BFF9A3: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5BF595A: GOMP\_parallel (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5270C2E: ggml\_graph\_compute (ggml-cpu.c:3333)

==463887== by 0x5272636: ggml\_backend\_cpu\_graph\_compute(ggml\_backend\*, ggml\_cgraph\*) (ggml-cpu.cpp:190)

==463887== by 0x5545BE3: ggml\_backend\_graph\_compute\_async (ggml-backend.cpp:452)

==463887== by 0x554AD50: ggml\_backend\_sched\_compute\_splits(ggml\_backend\_sched\*) (ggml-backend.cpp:1678)

==463887== by 0x554BD1C: ggml\_backend\_sched\_graph\_compute\_async (ggml-backend.cpp:1901)

==463887== by 0x4DB0BDD: llama\_context::graph\_compute(ggml\_cgraph\*, bool) (llama-context.cpp:2326)

==463887== by 0x4DAB802: llama\_context::process\_ubatch(llama\_ubatch const&, llm\_graph\_type, llama\_memory\_context\_i\*, ggml\_status&) (llama-context.cpp:1309)

==463887== by 0x4DABC9A: llama\_context::encode(llama\_batch const&) (llama-context.cpp:1382)

==463887==

==463887== 960 bytes in 3 blocks are possibly lost in loss record 6 of 13

==463887== at 0x484D953: calloc (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x40145AB: calloc (rtld-malloc.h:44)

==463887== by 0x40145AB: allocate\_dtv (dl-tls.c:370)

==463887== by 0x40145AB: \_dl\_allocate\_tls (dl-tls.c:629)

==463887== by 0x5A66616: allocate\_stack (allocatetestack.c:429)

==463887== by 0x5A66616: pthread\_create@@GLIBC\_2.34 (pthread\_create.c:655)

==463887== by 0x5BFFE1E: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5BF5971: GOMP\_parallel (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5270C2E: ggml\_graph\_compute (ggml-cpu.c:3333)

==463887== by 0x5272636: ggml\_backend\_cpu\_graph\_compute(ggml\_backend\*, ggml\_cgraph\*) (ggml-cpu.cpp:190)

==463887== by 0x5545BE3: ggml\_backend\_graph\_compute\_async (ggml-backend.cpp:452)

==463887== by 0x554AD50: ggml\_backend\_sched\_compute\_splits(ggml\_backend\_sched\*) (ggml-backend.cpp:1678)

==463887== by 0x554BD1C: ggml\_backend\_sched\_graph\_compute\_async (ggml-backend.cpp:1901)

==463887== by 0x4DB0BDD: llama\_context::graph\_compute(ggml\_cgraph\*, bool) (llama-context.cpp:2326)

==463887== by 0x4DAB802: llama\_context::process\_ubatch(llama\_ubatch const&, llm\_graph\_type, llama\_memory\_context\_i\*, ggml\_status&) (llama-context.cpp:1309)

==463887==

==463887== 2,272 bytes in 1 blocks are still reachable in loss record 7 of 13

==463887== at 0x484DECC: memalign (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x5BEB550: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5BFF822: ??? (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5BF595A: GOMP\_parallel (in /usr/lib/x86\_64-linux-gnu/libgomp.so.1.0.0)

==463887== by 0x5270C2E: ggml\_graph\_compute (ggml-cpu.c:3333)

==463887== by 0x5272636: ggml\_backend\_cpu\_graph\_compute(ggml\_backend\*, ggml\_cgraph\*) (ggml-cpu.cpp:190)

==463887== by 0x5545BE3: ggml\_backend\_graph\_compute\_async (ggml-

backend.cpp:452)

==463887== by 0x554AD50:

ggml\_backend\_sched\_compute\_splits(ggml\_backend\_sched\*) (ggml-backend.cpp:1678)

==463887== by 0x554BD1C: ggml\_backend\_sched\_graph\_compute\_async (ggml-backend.cpp:1901)

==463887== by 0x4DB0BDD: llama\_context::graph\_compute(ggml\_cgraph\*, bool) (llama-context.cpp:2326)

==463887== by 0x4DAB802: llama\_context::process\_ubatch(llama\_ubatch const&, llm\_graph\_type, llama\_memory\_context\_i\*, ggml\_status&) (llama-context.cpp:1309)

==463887== by 0x4DABC9A: llama\_context::encode(llama\_batch const&) (llama-context.cpp:1382)

==463887==

==463887== 8,192 bytes in 1 blocks are still reachable in loss record 8 of 13

==463887== at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x575EBC3: std::basic\_filebuf<char, std::char\_traits<char> >::\_M\_allocate\_internal\_buffer() (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x575C8CA: ??? (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x5703B7B: std::ios\_base::sync\_with\_stdio(bool) (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x117804: main (main.cc:446)

==463887==

==463887== 8,192 bytes in 1 blocks are still reachable in loss record 9 of 13

==463887== at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x575EBC3: std::basic\_filebuf<char, std::char\_traits<char> >::\_M\_allocate\_internal\_buffer() (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x575C8CA: ??? (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x5703B9E: std::ios\_base::sync\_with\_stdio(bool) (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x117804: main (main.cc:446)

==463887==

==463887== 8,192 bytes in 1 blocks are still reachable in loss record 10 of 13

==463887== at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x575EBC3: std::basic\_filebuf<char, std::char\_traits<char> >::\_M\_allocate\_internal\_buffer() (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x575C8CA: ??? (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x5703BC1: std::ios\_base::sync\_with\_stdio(bool) (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x117804: main (main.cc:446)

==463887==

==463887== 32,768 bytes in 1 blocks are still reachable in loss record 11 of 13

==463887== at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x5760A56: std::basic\_filebuf<wchar\_t, std::char\_traits<wchar\_t> >::\_M\_allocate\_internal\_buffer() (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x575CACA: ??? (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x5703C30: std::ios\_base::sync\_with\_stdio(bool) (in /usr/lib/x86\_64-linux-gnu/libstdc++.so.6.0.33)

==463887== by 0x117804: main (main.cc:446)

==463887==

==463887== 32,768 bytes in 1 blocks are still reachable in loss record 12 of 13

==463887== at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload\_memcheck-amd64-linux.so)

==463887== by 0x5760A56: std::basic\_filebuf<wchar\_t,

```

std::char_traits<wchar_t> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-
gnu/libstdc++.so.6.0.33)

==463887==      by 0x575CACA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)

==463887==      by 0x5703C53: std::ios_base::sync_with_stdio(bool) (in
/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)

==463887==      by 0x117804: main (main.cc:446)

==463887==

==463887== 32,768 bytes in 1 blocks are still reachable in loss record 13 of 13

==463887==      at 0x48485C3: operator new[](unsigned long) (in
/usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)

==463887==      by 0x5760A56: std::basic_filebuf<wchar_t,
std::char_traits<wchar_t> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-
gnu/libstdc++.so.6.0.33)

==463887==      by 0x575CACA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)

==463887==      by 0x5703C76: std::ios_base::sync_with_stdio(bool) (in
/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)

==463887==      by 0x117804: main (main.cc:446)

==463887==

==463887== LEAK SUMMARY:

==463887==      definitely lost: 0 bytes in 0 blocks

==463887==      indirectly lost: 0 bytes in 0 blocks

==463887==      possibly lost: 960 bytes in 3 blocks

==463887==      still reachable: 125,496 bytes in 12 blocks

==463887==      suppressed: 0 bytes in 0 blocks

==463887==

==463887== For lists of detected and suppressed errors, rerun with: -s

==463887== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)

```

```

labuser@labuser:~/CPP_Project$ head -n 5 ./data/三国演义.txt > ./data/test_tiny.txt
labuser@labuser:~/CPP_Project$ valgrind --leak-check=full --show-leak-kinds=all \
./build_linux/mrag --ingest ./data/test_tiny.txt ./data/test_tiny.db
==463887== Memcheck, a memory error detector
==463887== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==463887== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==463887== Command: ./build_linux/mrag --ingest ./data/test_tiny.txt ./data/test_tiny.db
==463887==
[INFO] 正在处理文档: ./data/test_tiny.txt
[INFO] 当前文档切分完成, 共 2 个片段
[INFO] 全部文档切分完成, 共 2 个片段
[INFO] 正在加载 Embedding 模型: ./models/bge-small-zh-v1.5-f16.gguf
[INFO] 正在生成向量嵌入...
[INFO] 嵌入进度: 2/2
[INFO] 正在写入向量数据库...
[INFO] 知识库构建完成, 保存至: ./data/test_tiny.db
[INFO] 数据库包含 2 个向量条目
==463887==
==463887== HEAP SUMMARY:
==463887==      in use at exit: 126,456 bytes in 15 blocks
==463887==    total heap usage: 28,480 allocs, 28,465 frees, 46,596,058 bytes allocated
==463887==
==463887== 8 bytes in 1 blocks are still reachable in loss record 1 of 13
==463887== at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887== by 0x5BE84A4: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5C0210E: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5B89C30: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x400571E: call_init.part.0 (dl-init.c:74)
==463887== by 0x4005823: call_init (dl-init.c:120)
==463887== by 0x4005823: _dl_init (dl-init.c:121)
==463887== by 0x401F59F: ??? (in /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2)
==463887== by 0x3: ???
==463887== by 0x1FFEFFFE02: ???
==463887== by 0x1FFEFFFE15: ???
==463887== by 0x1FFEFFFE1E: ???
==463887== by 0x1FFEFFFE33: ???
==463887==
==463887== 8 bytes in 1 blocks are still reachable in loss record 2 of 13
==463887== at 0x4846A3F: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887== by 0x528A530: ggml_backend_cpu_repack_buffer_type() (repack.cpp:4832)
==463887== by 0x5271F1A: ggml_backend_cpu_get_extra_buffer_types()::(lambda)#1::operator()() const (ggml-cpu.cpp:65)
==463887== by 0x5271FF5: ggml_backend_cpu_get_extra_buffer_types() (ggml-cpu.cpp:71)
==463887== by 0x527268C: ggml_backend_cpu_device_get_extra_buffers_type(ggml_backend_device*)::(lambda)#1::operator()() const (ggml-cpu.cpp:78)
==463887== by 0x5272191: ggml_backend_cpu_device_get_extra_buffers_type(ggml_backend_device*) (ggml-cpu.cpp:81)
==463887== by 0x4E8E820: make_cpu_buf_list(std::vector<llama_device, std::allocator<llama_device>> const&, bool, bool) (llama-model.cpp:861)
==463887== by 0x4EC00E9: llama_model_base::load_tensors(llvm_model_loader&) (llama-model.cpp:1178)
==463887== by 0x4D56ECE: llama_model_load(gguf_context*, void (*)(ggml_tensor*, void*), void*, std::cxx11::basic_string<char, std::char_traits<char>, std::allocator<char>> const&, std::vector<std::cxx11::basic_string<char, std::char_traits<char>, std::allocator<char>> const&, bool, bool>, llama_model_params&) (llama.cpp:327)
==463887== by 0x4D5EE4F: llama_model_load_from_file_impl(gguf_context*, void (*)(ggml_tensor*, void*), void*, std::cxx11::basic_string<char, std::char_traits<char>, std::allocator<char>> const&, std::vector<std::cxx11::basic_string<char, std::char_traits<char>, std::allocator<char>> const&, bool, bool>, llama_model_params&) (llama.cpp:386)
==463887== by 0x4D5EE4F: llama_model_load_from_file_impl(gguf_context*, void (*)(ggml_tensor*, void*), void*, std::cxx11::basic_string<char, std::char_traits<char>, std::allocator<char>> const&, std::vector<std::cxx11::basic_string<char, std::char_traits<char>, std::allocator<char>> const&, bool, bool>, llama_model_params&) (llama.cpp:386)
labuser@labuser:~/CPP_Project$ valgrind --leak-check=full --show-leak-kinds=all \
./build_linux/mrag --ingest ./data/test_tiny.txt ./data/test_tiny.db
==463887== by 0x13D44: llama_model_base::llama_model_base(std::cxx11::basic_string<char, std::char_traits<char>, std::allocator<char>> const&, ModelMode, AppConfig::GenerationConfig const&, AppConfig::EmbeddingConfig const&, int) (llama_model.cc:18)
==463887==
==463887== 40 bytes in 1 blocks are still reachable in loss record 3 of 13
==463887== at 0x484D880: realloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887== by 0x5BE8B14: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5C0210E: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5B89C30: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5270C2E: ggml_graph_compute (ggml-cpu.c:3333)
==463887== by 0x5272636: ggml_backend_cpu_graph_compute(ggml_backend*, ggml_cgraph*) (ggml-cpu.cpp:190)
==463887== by 0x55458E3: ggml_backend_graph_compute_async (ggml-backend.cpp:452)
==463887== by 0x554AD50: ggml_backend_sched_compute_splits(ggml_backend_sched*) (ggml-backend.cpp:1678)
==463887== by 0x5548D1C: ggml_backend_sched_graph_compute_async (ggml-backend.cpp:1901)
==463887== by 0x4D08B00: llama_context::graph_compute(ggml_cgraph*, bool) (llama-context.cpp:2326)
==463887== by 0x4D48802: llama_context::process_ubatch(llvm_ubatch const&, llm_graph_type, llama_memory_context_i*, ggml_status&) (llama-context.cpp:1309)
==463887== by 0x4D48C9A: llama_context::encode(llvm_batch const&) (llama-context.cpp:1382)
==463887==
==463887== 96 bytes in 1 blocks are still reachable in loss record 4 of 13
==463887== at 0x484D953: calloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887== by 0x5BE84DC: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5BE9BA0: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5BE9C77: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x400571E: call_init.part.0 (dl-init.c:74)
==463887== by 0x4005823: call_init (dl-init.c:120)
==463887== by 0x4005823: _dl_init (dl-init.c:121)
==463887== by 0x401F59F: ??? (in /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2)
==463887== by 0x3: ???
==463887== by 0x1FFEFFFE02: ???
==463887== by 0x1FFEFFFE15: ???
==463887== by 0x1FFEFFFE1E: ???
==463887== by 0x1FFEFFFE33: ???
==463887==
==463887== 192 bytes in 1 blocks are still reachable in loss record 5 of 13
==463887== at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887== by 0x5BE84A4: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5B8F9A3: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5B8F95A: GOMP_parallel (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887== by 0x5270C2E: ggml_graph_compute (ggml-cpu.c:3333)
==463887== by 0x5272636: ggml_backend_cpu_graph_compute(ggml_backend*, ggml_cgraph*) (ggml-cpu.cpp:190)
==463887== by 0x55458E3: ggml_backend_graph_compute_async (ggml-backend.cpp:452)
==463887== by 0x554AD50: ggml_backend_sched_compute_splits(ggml_backend_sched*) (ggml-backend.cpp:1678)
==463887== by 0x5548D1C: ggml_backend_sched_graph_compute_async (ggml-backend.cpp:1901)
==463887== by 0x4D08B00: llama_context::graph_compute(ggml_cgraph*, bool) (llama-context.cpp:2326)
==463887== by 0x4D48802: llama_context::process_ubatch(llvm_ubatch const&, llm_graph_type, llama_memory_context_i*, ggml_status&) (llama-context.cpp:1309)
==463887== by 0x4D48C9A: llama_context::encode(llvm_batch const&) (llama-context.cpp:1382)
==463887==
==463887== 960 bytes in 3 blocks are possibly lost in loss record 6 of 13
==463887== at 0x484D953: calloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887== by 0x40145A8: calloc (rld-malloc.h:44)
==463887== by 0x40145A8: allocate_dtv (dl-tls.c:370)
==463887== by 0x40145A8: _dl_allocate_tls (dl-tls.c:629)

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labuser@labuser:~/CPP_Project$ valgrind --leak-check=full --show-leak-kinds=all \
./build_linux/mrag --ingest ./data/test_tiny.txt ./data/test_tiny.db
==463887== 960 bytes in 3 blocks are possibly lost in loss record 6 of 13
==463887==    at 0x484D953: calloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x40145A8: calloc (rtld-malloc.h:44)
==463887==    by 0x40145A8: allocate_dtv (dl-tls.c:378)
==463887==    by 0x40145A8: _dl_allocate_tls (dl-tls.c:629)
==463887==    by 0x5A66616: allocate_stack (allocatestack.c:429)
==463887==    by 0x5A66616: pthread_create@@GLIBC_2.34 (pthread_create.c:655)
==463887==    by 0x58FFE1E: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==    by 0x58F5971: GOMP_parallel (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==    by 0x52726C2: ggml_graph_compute (ggml-cpu.c:3333)
==463887==    by 0x5272636: ggml_backend_cpu_graph_compute(ggml_backend*, ggml_cgraph*) (ggml-cpu.cpp:198)
==463887==    by 0x55458E3: ggml_backend_graph_compute_async (ggml-backend.cpp:452)
==463887==    by 0x554A050: ggml_backend_sched_compute_splits(ggml_backend_sched*) (ggml-backend.cpp:1678)
==463887==    by 0x5548D1C: ggml_backend_sched_graph_compute_async (ggml-backend.cpp:1901)
==463887==    by 0x4D898D0: llama_context::graph_compute(ggml_cgraph*, bool) (llama-context.cpp:2326)
==463887==    by 0x4D88802: llama_context::process_ubatch(llama_ubatch const&, llm_graph_type, llama_memory_context_i*, ggml_status&) (llama-context.cpp:1309)
==463887==
==463887== 2,272 bytes in 1 blocks are still reachable in loss record 7 of 13
==463887==    at 0x484DECC: memalign (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x58EB550: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==    by 0x58FF822: ??? (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==    by 0x58F595A: GOMP_parallel (in /usr/lib/x86_64-linux-gnu/libgomp.so.1.0.0)
==463887==    by 0x52726C2: ggml_graph_compute (ggml-cpu.c:3333)
==463887==    by 0x5272636: ggml_backend_cpu_graph_compute(ggml_backend*, ggml_cgraph*) (ggml-cpu.cpp:198)
==463887==    by 0x55458E3: ggml_backend_graph_compute_async (ggml-backend.cpp:452)
==463887==    by 0x554A050: ggml_backend_sched_compute_splits(ggml_backend_sched*) (ggml-backend.cpp:1678)
==463887==    by 0x5548D1C: ggml_backend_sched_graph_compute_async (ggml-backend.cpp:1901)
==463887==    by 0x4D898D0: llama_context::graph_compute(ggml_cgraph*, bool) (llama-context.cpp:2326)
==463887==    by 0x4D88802: llama_context::process_ubatch(llama_ubatch const&, llm_graph_type, llama_memory_context_i*, ggml_status&) (llama-context.cpp:1309)
==463887==    by 0x4D8C9A: llama_context::encode(llama_batch const&) (llama-context.cpp:1382)
==463887==
==463887== 8,192 bytes in 1 blocks are still reachable in loss record 8 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759BC3: std::basic_filebuf<char, std::char_traits<char> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759BCA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783B78: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== 8,192 bytes in 1 blocks are still reachable in loss record 9 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759BC3: std::basic_filebuf<char, std::char_traits<char> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759BCA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783B78: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== 8,192 bytes in 1 blocks are still reachable in loss record 10 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759BC3: std::basic_filebuf<char, std::char_traits<char> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759BCA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783B78: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)

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labuser@labuser:~/CPP_Project$ valgrind --leak-check=full --show-leak-kinds=all \
./build_linux/mrag --ingest ./data/test_tiny.txt ./data/test_tiny.db
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759BC3: std::basic_filebuf<char, std::char_traits<char> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759BCA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783B78: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== 8,192 bytes in 1 blocks are still reachable in loss record 9 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759BC3: std::basic_filebuf<char, std::char_traits<char> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759BCA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783B78: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== 8,192 bytes in 1 blocks are still reachable in loss record 10 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759BC3: std::basic_filebuf<char, std::char_traits<char> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759BCA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783B78: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== 32,768 bytes in 1 blocks are still reachable in loss record 11 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759A56: std::basic_filebuf<wchar_t, std::char_traits<wchar_t> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759ACA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783C30: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== 32,768 bytes in 1 blocks are still reachable in loss record 12 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759A56: std::basic_filebuf<wchar_t, std::char_traits<wchar_t> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759ACA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783C30: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== 32,768 bytes in 1 blocks are still reachable in loss record 13 of 13
==463887==    at 0x48485C3: operator new[](unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==463887==    by 0x5759A56: std::basic_filebuf<wchar_t, std::char_traits<wchar_t> >::_M_allocate_internal_buffer() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5759ACA: ??? (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x5783C76: std::ios_base::sync_with_stdio(bool) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==463887==    by 0x117804: main (main.cc:446)
==463887==
==463887== LEAK SUMMARY:
==463887==    definitely lost: 0 bytes in 0 blocks
==463887==    indirectly lost: 0 bytes in 0 blocks
==463887==    possibly lost: 960 bytes in 3 blocks
==463887==    still reachable: 125,496 bytes in 12 blocks
==463887==    suppressed: 0 bytes in 0 blocks
==463887==
==463887== For lists of detected and suppressed errors, rerun with: -s
==463887== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
labuser@labuser:~/CPP_Project$

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